This article has been accepted for publication in IEEE Transactions on Vehicular Technology. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVT.2026.3706538

Beam Squint Assisted Joint Angle-Distance Localization for Near-Field Communications

Aibiao Zhang, *Graduate Student Member, IEEE,* Weizheng Zhang, *Member, IEEE,* Chiya Zhang, *Member, IEEE,* and Tingting Zhang, *Member, IEEE,*

***Abstract*—The advent of extremely large-scale MIMO (XL-MIMO) provides crucial support for future sixth-generation (6G) systems to achieve high-precision, low-latency user localization. In near-field wideband scenarios, conventional beamforming based on frequency-independent phase shifters (PS) leads to a severe beam-squint phenomenon, where beams on different subcarriers point to inconsistent directions, thereby significantly degrading localization accuracy. However, by combining true-time-delay (TTD) elements and PS, the beam-squint effect can be exploited to establish a controllable mapping between subcarriers and spatial locations for localization purposes. Yet, most existing near-field localization methods adopt a two-step strategy that first estimates the angle and then the distance. This sequential process is susceptible to error propagation, impairing overall localization accuracy. To address these issues, in this paper we propose a beam-squint-assisted joint angle-distance (JAD) localization scheme with a coarse-to-fine two-stage estimator. The first stage rapidly obtains a coarse estimate of the user's position from the peak of the received power spectrum. The second stage applies a near-field improved multiple signal classification (MUSIC) algorithm to conduct a fine, high-resolution search around the coarse estimate, yielding a precise JAD estimate. Simulation results demonstrate that, compared with conventional two-step localization methods, the proposed method achieves significant improvements in localization accuracy and robustness in both single-user and multi-user near-field scenarios, highlighting its potential for future 6G sensing and localization systems.**

***Index Terms*—XL-MIMO, Near-field localization, beam squint, true-time-delay, MUSIC algorithm.**

I. Introduction

WITH the commercialization of fifth-generation (5G) networks, global research has shifted toward sixth-generation (6G) systems. Compared with 5G, 6G targets more stringent requirements, including centimeter- or even millimeter-level positioning accuracy to enable seamless localization in both indoor and outdoor scenarios. Promising enablers include extremely large-scale multiple-input multiple-output (XL-MIMO) systems [1], millimeter-wave/terahertz (mmWave/THz) communications , and ultra-wideband transmission. Unlike conventional far-field massive MIMO systems, where electromagnetic (EM) propagation is typically modeled by plane waves, XL-MIMO especially in high-frequency bands operates in the near-field region [2]. This shift necessitates spherical wave channel modeling rather than traditional plane wave assumptions. The spherical wave characteristics allow beamforming to focus energy on specific spatial locations, offering strong potential for high precision user localization [3], [4], [5].

The authors are with the School of Information Science and Technology, Harbin Institute of Technology, Shenzhen 518055, China (e-mail: 24s152040@stu.hit.edu.cn).

When users are located in the far-field region of the base station, early studies mainly focused on angle estimation, which can be formulated as a direction of arrival (DOA) estimation problem [6]. In array signal processing, a number of classical DOA estimation techniques have been developed. For instance, the multiple signal classification (MUSIC) algorithm, which exploits eigen-decomposition of the covariance matrix, serves as a cornerstone in spatial spectrum estimation and direction finding [7].

While far-field studies have primarily focused on DOA estimation, near-field scenarios present fundamentally different challenges and opportunities [8]. In the near-field region, the user's distance information becomes equally critical. When users are located within this region, electromagnetic (EM) waves must be accurately modeled as spherical waves [9]. The boundary between the far-field and near-field regions is determined by the Rayleigh distance, which is proportional to both the square of the array aperture and the carrier frequency [10]. With the enlarged aperture and increased frequency in broadband XL-MIMO systems, the near-field region can extend to several tens or even hundreds of meters [11]. Consequently, in XL-MIMO, beam focusing should be achieved by exploiting spherical wave characteristics to concentrate signals on specific spatial positions, rather than merely steering them toward angular directions as in conventional far-field beamforming [12]–[15]. For example, H. Hua investigated integrated sensing and communication (ISAC) systems in the near-field, where echo signals were simultaneously utilized for both communication and localization [13]. In [14], the performance of several tensor decomposition based three dimensional (3D) localization estimators was compared. Furthermore, a Bayesian tracking algorithm was developed in [15] to track a single near-field target in 3D space. [16] proposes and verifies two efficient near-field beam training schemes based on the conventional discrete Fourier transform (DFT).

A. Related Works

For XL-MIMO systems, extremely large bandwidths introduce the beam squint effect, where beams at different frequencies focus on different spatial locations [17]. As a result, many frequency-dependent beams may not align with the true user position, leading to significant array gain loss [18]. However, this phenomenon can be transformed into an asset for localization. By carefully designing true-time-delay (TTD) and phase shifter (PS) parameters, it is possible to create a controllable rainbow beam establishing a one-to-one mapping between subcarriers and spatial locations [19]. This principle has been actively explored for near-field localization. For instance, H. Luo pioneered a method that jointly optimized PS and TTDs to form a controllable beam trajectory, enabling direct angle-distance localization from power peaks [20]. While innovative, their approach relies on a sequential, two-step estimation strategy: first finding the angle, then the distance. This separation is a critical weakness, as it is highly susceptible to error propagation, where inaccuracies in the initial angle estimate degrade the subsequent distance estimation. Building on this, [21] integrated a CNN with controllable squint beams to address spatial non-stationarity, but this still operates within a beam training framework and does not fundamentally solve the error propagation problem. In a different approach, A. Gast proposed a deep-learning-aided cascaded MUSIC algorithm; however, its major limitation is being designed for narrowband systems, meaning it fails to address the beam squint effect inherent in the wideband scenarios we consider [22].

B. Motivations and Contributions

To the best of our knowledge, existing near-field localization studies have not yet established a joint angle-distance framework that explicitly avoids the error propagation inherent in conventional angle-first-then-distance schemes. To fill this gap, we propose a beam-squint-assisted coarse-to-fine MUSIC framework for near-field joint angle-distance (JAD) localization, where a coarse joint estimate is first obtained through beam-squint-assisted probing and then refined via a localized near-field MUSIC search. By avoiding sequential decoupled estimation, the proposed framework mitigates error propagation while maintaining high-resolution localization capability. Our main contributions are summarized as follows:

- We develop a beam-squint-assisted controllable beam pattern design for near-field JAD localization. By jointly configuring the TTD units and phase shifters, a one-to-one frequency-space mapping between subcarriers and target spatial coordinates is established over the sensing region. We further clarify the monotonicity and uniqueness conditions of this mapping, which provide the foundation for unambiguous coarse joint localization.
- Based on the above beam pattern, we propose a coarse-to-fine joint localization algorithm. In the first stage, a coarse angle-distance estimate is directly obtained from the beam-squint-assisted power spectrum. In the second stage, a localized near-field MUSIC refinement is performed around the coarse estimate, where near-field geometry-compensated spatial smoothing, eigenvalue decomposition, and multi-subcarrier geometric averaging are jointly exploited to improve accuracy and robustness.
- We conduct extensive simulations and comparative evaluations to validate the proposed method. Results show that, compared with conventional two-step schemes and power-peak baselines, our approach achieves superior localization accuracy within a single scan, with an angle root mean square error (RMSE) below 0.001° and a distance RMSE below 0.001 m under the considered settings. The method also demonstrates strong two-dimensional resolution in both single-user and multi-user scenarios, and remains robust even under severe near-field beam-squint effects in wideband systems.

C. Organization and Notation

The remainder of this paper is organized as follows. Section II introduces the system model and near-field channel modeling. Section III analyzes the near-field beam squint phenomenon and reviews the controllable beam squint strategy enabled by TTDs. In Section IV, we propose a joint angle distance localization scheme based on the MUSIC algorithm with beam squint assistance. Section V and Section VI present the simulation results and discussions, respectively.

The notations used throughout this paper are summarized as follows. Scalars are denoted by italic letters, while vectors and matrices are represented by bold lowercase and bold uppercase letters, respectively. For a matrix $\mathbf{A}$, $\mathbf{A}^T$, $\mathbf{A}^H$, and $\mathbf{A}^{-1}$ denote its transpose, conjugate transpose, and inverse, respectively. The notation $|\cdot|$ represents the absolute value of a scalar or the determinant of a matrix, while $||\cdot||$ denotes the Euclidean norm of a vector. The operator $\mathbb{E}\cdot$ stands for statistical expectation, and $\Re\cdot$ denotes the real part. The symbols $\otimes$, $\odot$, and $\circ$ denote the Kronecker product, Hadamard product, and outer product of vectors, respectively.

II. System Model

In this work, we consider a large-scale MIMO scenario operating in the wideband mmWave regime. An N-antenna uniform linear array (ULA) with a single radio-frequency (RF) chain architecture is deployed at the BS to support K users. The transmit signals employ OFDM modulation with bandwidth W. For the OFDM system with M subcarriers, the subcarrier index is defined as $m \in \{0, 1, \ldots, M-1\}$, and the subcarrier spacing is given by $\Delta f = \frac{W}{M}$. Accordingly, the frequency of the m-th subcarrier is expressed as

$$f_m = f_c - \frac{W}{2} + m\Delta f, \quad m = 0, 1, \ldots, M-1, \tag{1}$$

where f_c denotes the carrier (center) frequency. For notational clarity, we use f_c to denote the carrier frequency throughout the manuscript, while $\widetilde{f}_m = f_m - f_c$ denotes the frequency offset of the m-th subcarrier from the carrier frequency. The k-th user is located at Cartesian coordinates (x_k, y_k), which correspond to polar coordinates (r_k, θ_k).

In the mmWave and terahertz regimes, the dominant component is typically LoS link, and we adopt a single-path model for tractability, while the framework remains extensible to multipath scenarios. Given the large array aperture D and short wavelength λ, the near-field region delimited by the Rayleigh distance $Z = \frac{2D^2}{\lambda}$ becomes the primary operational zone. To accurately characterize the spherical wavefronts within this region, the distance $r_{k,n}$ between the k-th user (target) and the n-th BS antenna is formulated using the second-order Fresnel

This article has been accepted for publication in IEEE Transactions on Vehicular Technology. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVT.2026.3706538

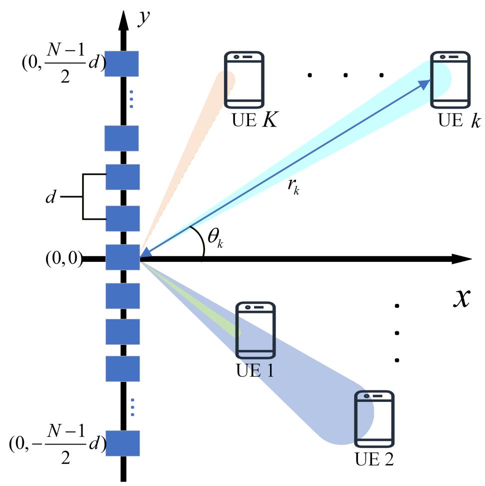


Fig. 1. Near-field system model.

approximation, following the near-field modeling framework in [20]:

$$\begin{aligned} r_{k,n} &= \sqrt{x_k^2 + (y_k - nd)^2} = \sqrt{r_k^2 + n^2 d^2 - 2 r_k n d \sin\theta_k} \\ &\approx r_k - nd\sin\theta_k + \frac{n^2 d^2 \cos^2\theta_k}{2 r_k}. \end{aligned} \tag{2}$$

To characterize the multi-user scenario, we stack the channel vectors of all K users to form the aggregate channel matrix $\mathbf{H}_m \in \mathbb{C}^{N\times K}$ for the m-th subcarrier:

$$\mathbf{H}_m = [\mathbf{h}_{1,m}, \mathbf{h}_{2,m}, \ldots, \mathbf{h}_{K,m}]. \tag{3}$$

The channel vector for the k-th user at the m-th subcarrier, denoted as $\mathbf{h}_{k,m} \in \mathbb{C}^{N\times 1}$, is modeled as:

$$\mathbf{h}_{k,m} = \sqrt{N}\beta_{k,m}\mathbf{a}(r_k, \theta_k, f_m), \tag{4}$$

where $\beta_{k,m}$ represents the complex path loss gain. The vector $\mathbf{a}(r_k, \theta_k, f_m) \in \mathbb{C}^{N\times 1}$ is the location-dependent near-field array response vector. Based on the Fresnel approximation, the phase of the n-th element is determined by the propagation distance difference. Thus, the n-th entry of the near-field steering vector is given by:

$$\begin{aligned} &\mathbf{a}(r_k, \theta_k, f_m) \\ &= \frac{1}{\sqrt{N}}\left[e^{-j\frac{2\pi f_m}{c} r_{k,\frac{-(N-1)}{2}}}, \ldots, e^{-j\frac{2\pi f_m}{c} r_{k,\frac{(N-1)}{2}}}\right]. \end{aligned} \tag{5}$$

is the near-field array response vector.

Given the single RF chain architecture, the BS implements analog beamforming via a network of phase shifters (PSs), which imposes a constant modulus constraint on the precoder. To focus the probing energy onto a specific reference location (r_0, θ_0), the beamforming vector $\mathbf{w}(r_0, \theta_0) \in \mathbb{C}^{N\times 1}$ is constructed by extracting the phase information of the array response at the reference frequency f_0. Consequently, the phase configuration for the n-th antenna element is formulated as

$$[w(r_0, \theta_0)]_n = \frac{1}{\sqrt{N}} e^{-j2\pi f_0 \frac{r_{0,n}}{c}}, \tag{6}$$

where $r_{0,n}$ denotes the distance between the reference focusing location (r_0, θ_0) and the n-th antenna element of the array.

During the localization phase, the BS transmits known pilot symbols using the analog beamforming vector $\mathbf{w}(r_0, \theta_0)$. Under the monostatic single-path assumption, the array-level received echo signal associated with user k at subcarrier m before receive combining is modeled as

$$\mathbf{r}_{k,m} = \gamma_{k,m}\mathbf{h}_{k,m} + \mathbf{n}_{k,m}, \qquad \gamma_{k,m} = \mathbf{h}_{k,m}^H \mathbf{w}\, s_{k,m}, \tag{7}$$

where $\gamma_{k,m} \in \mathbb{C}$ denotes the effective echo coefficient including the transmit beamforming gain and pilot symbol, and $\mathbf{n}_{k,m} \in \mathbb{C}^{N\times 1}$ is the array noise vector.

Under the single-RF-chain architecture, during the t-th probing interval the BS applies an analog combining vector $\mathbf{v}_t \in \mathbb{C}^{N\times 1}$, and the corresponding scalar observation is

$$y_{k,m}^{(t)} = \mathbf{v}_t^H \mathbf{r}_{k,m} = \mathbf{v}_t^H \gamma_{k,m}\mathbf{h}_{k,m} + \mathbf{v}_t^H \mathbf{n}_{k,m}, \tag{8}$$

where $y_{k,m}^{(t)} \in \mathbb{C}$.

By collecting T sequential measurements, we obtain the stacked observation

$$\tilde{\mathbf{y}}_{k,m} = \begin{bmatrix} y_{k,m}^{(1)} & y_{k,m}^{(2)} & \cdots & y_{k,m}^{(T)} \end{bmatrix}^T = \mathbf{V}^H \mathbf{r}_{k,m}, \tag{9}$$

with

$$\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_T \end{bmatrix} \in \mathbb{C}^{N\times T}. \tag{10}$$

When $T = N$ and $\mathbf{V} = \mathbf{I}_N$, the effective array-level observation is reconstructed as

$$\tilde{\mathbf{y}}_{k,m} = \mathbf{r}_{k,m} \triangleq \mathbf{y}_{k,m} \in \mathbb{C}^{N\times 1}, \tag{11}$$

which is the spatial snapshot used in the subsequent MUSIC-based refinement. In the multi-user case, the observations for different users can be separated by orthogonal pilots or scheduling in the time-frequency-code domain, enabling user-wise localization.

III. Near Field Controllable Beam Squint

In this section, we first analyze the beam squint effect in near-field communications. Then, we will briefly review the controllable beam squint control method proposed in [20].

A. Near-Field Beam Squint Effect

While the beamforming vector $\mathbf{w}(r_0, \theta_0)$ is designed to focus energy at a reference location (r_0, θ_0) for the central frequency f_0, the mismatch in phase responses across the wide bandwidth causes the beam focus of other subcarriers f_m to deviate from the target.

The array gain is defined as:

$$\begin{aligned} g(r, \theta, f_m, \mathbf{w}) &= \left|\mathbf{w}^H \cdot \mathbf{a}(r, \theta, f_m)\right| \\ &= \frac{1}{N}\left|\sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} e^{j2\pi f_0 \frac{r_{0,n}}{c}} e^{-j2\pi f_m \frac{r_n}{c}}\right| \\ &\approx \frac{1}{N}\left|\sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} \begin{matrix} e^{j\frac{2\pi n^2 d^2}{c}\left(f_m \frac{\cos^2\theta}{2r} - f_0 \frac{\cos^2\theta_0}{2r_0}\right)} \\ \times\, e^{-j\frac{2\pi n d}{c}(f_m \sin\theta - f_0 \sin\theta_0)} \end{matrix}\right|. \end{aligned} \tag{12}$$

Then, a feasible solution for (r, θ) maximzing the array gain can be written as

$$\sin\theta = \frac{f_0}{f_m}\sin\theta_0, \tag{13}$$

$$r = r_0 \cdot \frac{f_m}{f_0} \cdot \frac{\cos^2\theta}{\cos^2\theta_0}. \tag{14}$$

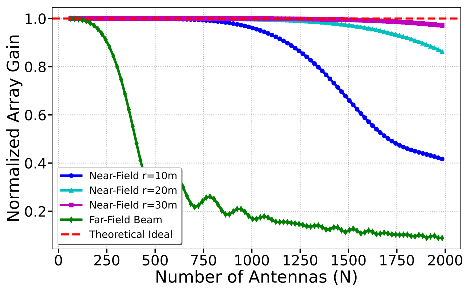


Fig. 2. Normalized array gain versus the number of antennas N.

Fig. 2 validates the accuracy of the beam focusing position (r_m, θ_m) derived in (13) and (14) by illustrating the normalized array gain versus the number of antennas N.

First, the simulation results confirm the effectiveness of the proposed model in typical near-field scenarios. For the considered XL-MIMO setup with $N = 256$, the Rayleigh distance extends to approximately $Z \approx 230$ m. The target range in the simulation ($r \approx 30$ m) falls deep within the radiating near-field (Fresnel) region. As observed in Fig. 2, the proposed near-field beam closely tracks the theoretical ideal within this range, maintaining a normalized gain > 0.998. This provides strong evidence that the second-order approximation in (2) is highly reliable for typical scenarios, and the derived trajectory in (13) and (14) accurately captures the beam focusing point. In contrast, the far-field beam suffers from severe degradation due to the mismatch of the spherical wavefront, underscoring the necessity of the proposed near-field model.

However, a distinct phenomenon is observed at the tail of the curves: when the number of antennas increases to an extremely large scale (e.g., $N > 1000$) and the target distance is extremely short (e.g., $r = 10$ m), the array gain begins to degrade. This indicates that the system enters the 'very near-field' region, where the second-order approximation in (2) begins to lose validity. To clarify the mechanism of this degradation, we analyze the discrepancy between the physical model in (12) and the approximation model in (2).

According to the definition of array gain in (12), the exact gain depends on the coherence between the designed phase and the true physical phase. The true phase is determined by the exact Euclidean distance:

$$r_{exact}(n) = \sqrt{r^2 + (nd)^2 - 2r(nd)\sin\theta}. \tag{15}$$

However, the beam trajectory derived in (13) and (14), as well as the array gain approximation in (12), are all predicated on the Fresnel approximation in (2):

$$r_{Fresnel}(n) \approx r - nd\sin\theta + \frac{(nd)^2\cos^2\theta}{2r}. \tag{16}$$

To quantify the residual error between the exact and approximate models, we perform a fourth-order Taylor expansion on $r_{exact}(n)$. Letting $x = nd$, the exact distance can be expanded as:

$$\begin{aligned} r_{\text{exact}} \approx\ & \underbrace{r - x\sin\theta + \frac{x^2\cos^2\theta}{2r}}_{\text{Second-order term in Eq. (1)}} \\ & + \underbrace{\frac{x^3\sin\theta\cos^2\theta}{2r^2}}_{\text{Third-order error}} \\ & - \underbrace{\frac{x^4(1-5\sin^2\theta)}{8r^3}}_{\text{Fourth-order error}} + \dots \end{aligned} \tag{17}$$

It is evident that (2) truncates terms of the third order and above. When the beam is steered towards the position determined by (13) and (14), the primary *residual phase error*, $\Delta\Phi(n)$, is dominated by the neglected higher-order terms:

$$\begin{aligned} \Delta\Phi(n) &= \frac{2\pi}{\lambda}\left(r_{\text{exact}} - r_{\text{Fresnel}}\right) \\ &\approx \frac{2\pi}{\lambda}\left(\frac{(nd)^3\mathcal{A}(\theta)}{2r^2} - \frac{(nd)^4\mathcal{B}(\theta)}{8r^3}\right), \end{aligned} \tag{18}$$

where $\mathcal{A}(\theta)$ and $\mathcal{B}(\theta)$ are angle-dependent coefficients.

Equation (18) reveals the mathematical root of the gain degradation: the phase error is proportional to $\frac{N^4}{r^3}$.

- As N increases, the error grows with the fourth power (N^4), indicating a rapid accumulation of phase mismatch at the array edges for XL-MIMO systems.
- As r decreases, the error grows with the inverse cube ($1/r^3$).

Consequently, in the extreme scenario where $r = 10$ m and N is massive, the accumulated residual phase error exceeds the beamforming tolerance, leading to the observed gain drop.

This phenomenon characterizes the validity boundary of the second-order approximation: the current model is accurate for typical near-field ranges, whereas extremely short ranges with ultra-large apertures may require higher-order or exact spherical-wave models.

Therefore, the near-field beam squint focusing point (r_m, θ_m) of the m-th subcarrier is determined by (13) and (14). Fig. 3 illustrates an example that clearly demonstrates the near-field beam-squint effect. As the subcarrier frequency f_m increases, the beamforming focus shifts to different spatial locations across subcarriers, and these focusing points form a continuous trajectory. The trajectory starts from the focus of the 0-th subcarrier at frequency f_0 and ends at that of the M-th subcarrier at frequency f_M. In Fig. 3, for instance, the beamforming focus at $f_0 = 27$ GHz is located at $(10\text{ m}, 60^\circ)$, while the focus at $f_c = 30$ GHz is $(17.44\text{ m}, 51.22^\circ)$, and the highest frequency subcarrier at $f_M = 33$ GHz is steered to $(24.34\text{ m}, 45.12^\circ)$. These results clearly show that the near-field beam squint effect becomes significant in wideband systems and cannot be neglected.

B. Controllable Beam Squint

Recently, several studies have focused on the beam squint effect in near-field XL-MIMO systems. A feasible solution

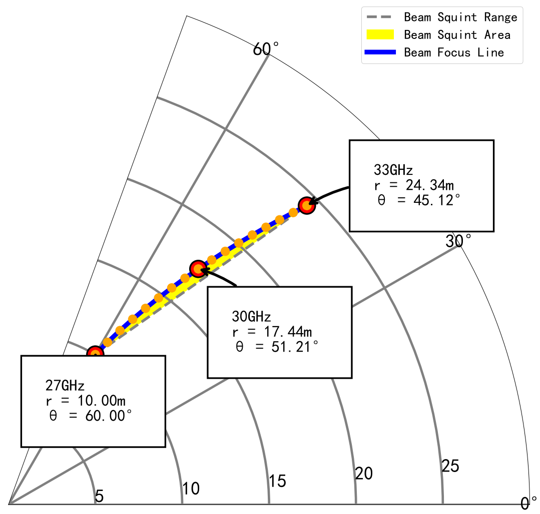


Fig. 3. Trajectory of near-field beam-squint focal points, where $f_c = 30$ GHz, $W = 6$ GHz, and $M = 16$. The beamforming focus of the lowest-frequency subcarrier is set to $(10\ \mathrm{m}, 60°)$, and the focal points of different subcarriers form a continuous trajectory in the angle-range domain due to the near-field beam-squint effect.

to this problem is to adopt beamforming based on TTD units rather than relying solely on PS. By leveraging TTDs, frequency dependent beams can be generated, thereby enabling controllable near-field beam squint..

In [20], this principle was leveraged to perform localization using a sequential, two-scan strategy. First, an angular scanning trajectory is designed to find a peak power subcarrier, which yields an initial angle estimate, $\hat{\theta}$. Subsequently, this angle estimate θ_m is treated as a fixed, known value to construct a second, radial scanning trajectory to determine the distance, r_m. The controllable focusing position can be expressed as

$$\sin\hat{\theta}_m = \frac{\left(W - \tilde{f}_m\right) f_0}{W f_m} \sin\theta_s + \frac{(W + f_0)\,\tilde{f}_m}{W f_m} \sin\theta_e, \quad (19)$$

$$\frac{1}{\hat{r}_m} = \frac{1}{r_s} \frac{\left(W - \tilde{f}_m\right) f_0}{W f_m} \frac{\cos^2\theta_s}{\cos^2\hat{\theta}_m} + \frac{1}{r_e} \frac{(W + f_0)\,\tilde{f}_m}{W f_m} \frac{\cos^2\theta_e}{\cos^2\hat{\theta}_m}, \quad (20)$$

where θ_s denotes the starting angular boundary of the beam-squint scanning trajectory.θ_e represents the ending angle of the scanning trajectory.

However, this sequential procedure has a fundamental weakness: error propagation. The initial angle estimate will inevitably contain an error, $\Delta\theta$, such that $\theta_m = \theta_{\text{true}} + \Delta\theta$. This error is then carried directly into the second stage. To quantify this effect, we analyze the distance trajectory equation from (20), which shows that the distance focus r_m is highly dependent on the angle term $\cos^2\theta_m$.

To quantify this effect, we can approximate the resulting distance error, Δr, by applying a first-order Taylor expansion with respect to the angle error $\Delta\theta$. This yields the following relationship:

$$\Delta r \approx -2 r_{\text{true}} \tan(\theta_{\text{true}}) \Delta\theta. \quad (21)$$

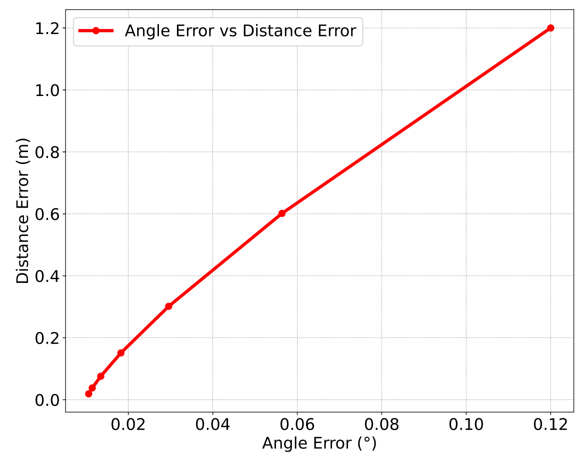


Fig. 4. The impact of error propagation in two-stage localization.

This approximation reveals a critical flaw: the final distance error Δr is not only proportional to the initial angular error $\Delta\theta$, but is also significantly amplified by the true distance r_{true}.

This analytical result is visualized in Fig. 4, which confirms that even minor angular inaccuracies are magnified into substantial distance errors, especially for targets at greater ranges. This inherent weakness degrades localization performance and motivates our work on a joint estimation scheme.

C. Beam Squint Assisted Beam Pattern Design

To overcome the beam squint limitation and achieve JAD localization, we propose a beam squint assisted beam pattern design that simultaneously optimizes the scanning paths in both the angular and radial domains. Unlike conventional narrowband beamforming, which focuses all subcarriers toward a single spatial point, the proposed method deliberately exploits the frequency-dependent focusing property to form a controlled beam pattern of subcarrier focal points in the near field. This beam pattern enables each subcarrier to illuminate a distinct position within the sensing region, establishing a one-to-one mapping between the subcarrier index and the spatial coordinates (θ, r). Such mapping provides the foundation for coarse joint angle–distance estimation in a single frequency sweep.

We define the sensing region of interest for the BS system as

$$\mathcal{S} = \{(\theta, r) \mid \theta_{\min} \le \theta \le \theta_{\max},\ r_{\min} \le r \le r_{\max}\}. \quad (22)$$

where the user is expected to be located. Two boundary points are selected to determine the localization endpoints: the starting focusing point $(\theta_s, r_s) = (\theta_{\min}, r_{\min})$ and the ending focusing point $(\theta_e, r_e) = (\theta_{\max}, r_{\max})$. These two points define the limits of the scanning coverage and ensure that all subcarriers collectively form a continuous pattern spanning the entire sensing region. In practice, the beam squint pattern is generated by jointly configuring the phase shifters and TTD elements according to equations (19) and (20). By assigning the lowest and highest subcarrier frequencies to the defined endpoints, the intermediate subcarriers naturally produce a smooth transition path between (θ_s, r_s) and (θ_e, r_e).

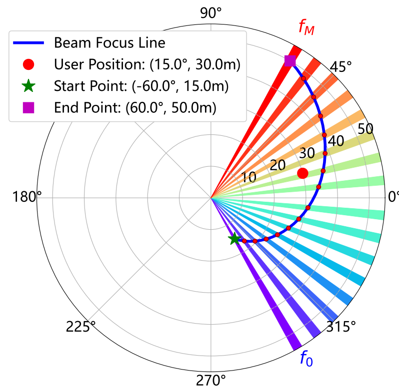


Fig. 5. Beam squint assisted beam pattern design. Where the beam pattern squints from $(-60^\circ, 15\text{m})$ to $(60^\circ, 50\text{m})$,the UE is located at $(15^\circ, 30\text{m})$.

As illustrated in Fig. 5, this design creates a beam pattern that squints continuously from $(-60^\circ, 15\,\text{m})$ to $(60^\circ, 50\,\text{m})$, scanning a wide area without mechanical steering. For this mapping to be unambiguous, the relationship between the subcarrier index m and the spatial coordinates (θ_m, r_m) must be monotonic. This is guaranteed by the structure of (19) and (20), ensuring that each subcarrier corresponds to a unique spatial point. This robust frequency-space correspondence is the key that enables our proposed joint localization scheme, which exploits this beam pattern to resolve angle and distance simultaneously, thereby avoiding the error accumulation inherent in multi-stage methods.

The following section introduces a beam squint assisted MUSIC algorithm, which exploits this correspondence to jointly estimate the user's position in a single scan, thereby avoiding the error accumulation inherent in multi-stage estimation schemes.

IV. Proposed Two Stage Localization

In this section, we propose a coarse-to-fine signal processing framework for near-field JAD localization. The first stage performs fast coarse localization by analyzing the power spectrum of the echo signal with low computational cost. Then, using the coarse estimate as prior information, a fine-grained two-dimensional spectral peak search is carried out within a small local region through a near-field MUSIC subspace algorithm with super-resolution capability, thereby refining the final joint estimate of angle and distance.

A. Coarse Localization via Power Spectrum

By configuring the parameters of the TTDs and PSs according to the designed JAD localization, the scalar observation used for coarse localization at the m-th subcarrier is denoted by $z_{k,m} \in \mathbb{C}$. The corresponding received power is expressed as

$$P_{k,m} = |z_{k,m}|^2. \tag{23}$$

If the user is located at (r_k, θ_k), its received power reaches the maximum at a specific subcarrier $f_{m^\star}$. We define the peak index as

$$m^\star = \arg\max_{m} P_{k,m}. \tag{24}$$

Then, according to the beam pattern design formulas (19) and (20), the coarse position estimate can be obtained as

$$(\hat{\theta}_k^{(0)}, \hat{r}_k^{(0)}) = (\tilde{\theta}_{m^\star}, \tilde{r}_{m^\star}). \tag{25}$$

This step does not require grid-based search, resulting in low computational complexity and making it suitable for fast coarse localization.

Although the coarse estimation algorithm is highly efficient, its performance is limited by two fundamental factors: (1) spatial quantization error, since the true target location usually lies between the discrete focusing points of the JAD localization; and (2) noise sensitivity, where low signal to noise ratio (SNR) conditions may cause the spectral peak to deviate. These limitations indicate that coarse estimation alone cannot meet the requirements of high accuracy localization. Therefore, it is essential to introduce a super-resolution algorithm that can surpass the Rayleigh limit and fully exploit the phase information of the array.

It should be noted that $z_{k,m}$ denotes the scalar observation used for coarse power-spectrum localization, whereas $\mathbf{y}_{k,m}$ denotes the reconstructed array-level snapshot used in the subsequent MUSIC-based refinement.

B. MUSIC Algorithm

MUSIC algorithm, as a representative algorithm for modern spectral estimation, is based on the subspace theory in linear algebra. For an array composed of N antenna elements, the covariance matrix $\mathbf{R}$ of the received signal can be decomposed into a set of orthogonal eigenvectors via eigenvalue decomposition (EVD). These eigenvectors span an N-dimensional signal space.

This orthogonality is the key to the super-resolution capability of the MUSIC algorithm. Specifically, for a steering vector $\mathbf{a}(\theta_k, r_k)$ that describes the true location of a target at (θ_k, r_k), it must satisfy:

$$\mathbf{a}(\theta_k, r_k)^H \mathbf{U}_n = \mathbf{0}. \tag{26}$$

Here, $\mathbf{U}_n$ is the matrix composed of the eigenvectors of the noise subspace. Therefore, we can construct a pseudo-spectrum function and search over all possible (θ, r). When the test point (θ, r) exactly corresponds to the true target location, its steering vector is orthogonal to the noise subspace, and a sharp peak with infinite value will appear in the pseudo-spectrum function.

As mentioned earlier, the key to applying the MUSIC algorithm in the near-field lies in using the correct near-field steering vector. This steering vector must accurately characterize the phase differences produced at each array element when a spherical wave impinges on the linear array. According to the Fresnel approximation, it can be expressed as

$$[\mathbf{a}(\theta, r, f_m)]_n = \left[e^{-j\frac{2\pi f_m}{c}\left(r - nd\sin\theta + \frac{n^2 d^2 \cos^2\theta}{2r}\right)} \right]. \tag{27}$$

This steering vector is a nonlinear function of both angle and distance. Compared with the far-field steering vector, it incorporates the distance dimension through the quadratic phase term $\frac{n^2 d^2 \cos^2\theta}{2r}$, which provides the physical foundation for achieving super-resolution estimation in the distance domain.

C. Fine Localization via MUSIC Algorithm

1) Single-Target Localization: After obtaining the coarse target location $(\hat{\theta}^{(0)}, \hat{r}^{(0)})$, the second stage performs a localized near-field MUSIC refinement within a small neighborhood around the coarse estimate, so as to achieve high-resolution joint angle-range localization with reduced complexity. For notational simplicity, the user index k is omitted in this subsection for the single-target case.

To further improve localization accuracy, local narrow-band MUSIC refinement is performed over several subcarriers around $m^\star$, e.g., $m^\star - \Delta m, \ldots, m^\star + \Delta m$. For a subcarrier f_m in this neighborhood, the reconstructed effective array-level snapshot obtained from sequential single-RF probing can be written as

$$\mathbf{y}_m = \alpha_m \mathbf{a}(\theta, r, f_m) + \mathbf{n}_m \in \mathbb{C}^{N\times 1}. \tag{28}$$

Its single-snapshot sample covariance matrix is

$$\hat{\mathbf{R}}_m = \mathbf{y}_m \mathbf{y}_m^H. \tag{29}$$

Since $\hat{\mathbf{R}}_m$ is rank-one, it cannot be directly used for stable subspace decomposition in MUSIC. Therefore, a subarray-based covariance construction is adopted. Different from conventional far-field spatial smoothing, where different shifted subarrays observe the same plane-wave structure up to a phase factor, the near-field spherical-wave steering vectors vary with the absolute antenna positions of different subarrays. Directly averaging the covariance matrices of different subarrays may therefore lead to a biased subspace estimate.

To address this issue, we introduce a near-field geometry-compensated spatial smoothing operation. The full array is divided into P overlapping subarrays, each containing M_s antennas. Let

$$\mathcal{N}_p = \{n_{p,1}, n_{p,2}, \ldots, n_{p,M_s}\} \tag{30}$$

denote the antenna index set of the p-th subarray. The received vector of the p-th subarray is denoted by $\mathbf{y}_m^{(p)} \in \mathbb{C}^{M_s\times 1}$. For a location parameter $\boldsymbol{\eta} = [\theta, r]^T$, the corresponding near-field subarray steering vector is written as

$$\mathbf{a}_m^{(p)}(\boldsymbol{\eta}) = \frac{1}{\sqrt{M_s}} \left[e^{-j\frac{2\pi f_m}{c} r_{n_{p,1}}(\boldsymbol{\eta})}, \ldots, e^{-j\frac{2\pi f_m}{c} r_{n_{p,M_s}}(\boldsymbol{\eta})} \right]^T, \tag{31}$$

where $r_n(\boldsymbol{\eta})$ denotes the distance between the target and the n-th antenna element under the adopted near-field model.

We select the central subarray p_0 as the reference subarray and use the first-stage coarse estimate

$$\hat{\boldsymbol{\eta}}^{(0)} = [\hat{\theta}^{(0)}, \hat{r}^{(0)}]^T \tag{32}$$

to construct the following phase-alignment matrix:

$$\mathbf{D}_{p,m}(\hat{\boldsymbol{\eta}}^{(0)}) = \mathrm{diag}\left\{ e^{-j\frac{2\pi f_m}{c}\left[r_{n_{p_0,q}}(\hat{\boldsymbol{\eta}}^{(0)}) - r_{n_{p,q}}(\hat{\boldsymbol{\eta}}^{(0)}) \right]} \right\}_{q=1}^{M_s}. \tag{33}$$

This matrix compensates for the deterministic near-field phase difference between the p-th subarray and the reference subarray. The phase-aligned subarray snapshot is obtained as

$$\tilde{\mathbf{y}}_m^{(p)} = \mathbf{D}_{p,m}(\hat{\boldsymbol{\eta}}^{(0)}) \mathbf{y}_m^{(p)}. \tag{34}$$

Then, the geometry-compensated smoothed covariance matrix is constructed as

$$\tilde{\mathbf{R}}_m = \frac{1}{P} \sum_{p=1}^{P} \tilde{\mathbf{y}}_m^{(p)} \left(\tilde{\mathbf{y}}_m^{(p)} \right)^H. \tag{35}$$

The rationale of this phase-alignment operation is as follows. If the coarse estimate is exact, i.e.,

$$\hat{\boldsymbol{\eta}}^{(0)} = \boldsymbol{\eta}, \tag{36}$$

then

$$\mathbf{D}_{p,m}(\boldsymbol{\eta}) \mathbf{a}_m^{(p)}(\boldsymbol{\eta}) = \mathbf{a}_m^{(p_0)}(\boldsymbol{\eta}), \quad \forall p. \tag{37}$$

Thus, all subarray signal components are mapped to the same reference-subarray steering vector after phase alignment. Consequently, the expected geometry-compensated covariance matrix has the form

$$\mathbb{E}[\tilde{\mathbf{R}}_m] = |\alpha_m|^2 \mathbf{a}_m^{(p_0)}(\boldsymbol{\eta}) \left(\mathbf{a}_m^{(p_0)}(\boldsymbol{\eta}) \right)^H + \sigma^2 \mathbf{I}, \tag{38}$$

which recovers the rank-one signal-plus-white-noise covariance structure required by MUSIC. Moreover, since $\mathbf{D}_{p,m}$ is a diagonal unit-modulus matrix, the white-noise covariance remains unchanged after phase alignment.

With a finite coarse-estimation error, the aligned covariance becomes a bounded perturbation of the ideal covariance. In the considered operating regime, the first-stage beam-squint-assisted coarse estimate is sufficiently accurate, and the local refinement window is selected to cover the coarse-estimation error. Therefore, the residual phase-alignment error remains within the perturbation bound and does not materially affect the final MUSIC refinement accuracy. Instead, the proposed phase-alignment operation removes the dominant subarray-dependent near-field phase mismatch and improves the validity of the subsequent subspace extraction.

After obtaining the geometry-compensated covariance matrix $\tilde{\mathbf{R}}_m$, eigenvalue decomposition (EVD) is performed as

$$\tilde{\mathbf{R}}_m = \tilde{\mathbf{E}}_m \tilde{\boldsymbol{\Lambda}}_m \tilde{\mathbf{E}}_m^H = \sum_{i=1}^{M_s} \tilde{\lambda}_{m,i} \tilde{\mathbf{e}}_{m,i} \tilde{\mathbf{e}}_{m,i}^H. \tag{39}$$

In the single-target scenario, the eigenvectors corresponding to the $M_s - 1$ smallest eigenvalues are used to construct the noise subspace matrix

$$\tilde{\mathbf{U}}_{n,m} = [\tilde{\mathbf{e}}_{m,2}, \ldots, \tilde{\mathbf{e}}_{m,M_s}]. \tag{40}$$

Since the aligned covariance matrix is referred to the reference subarray, the near-field MUSIC spectrum is evaluated using the reference-subarray steering vector:

$$P_{\mathrm{MUSIC},m}(\theta, r) = \frac{1}{\left(\mathbf{a}_m^{(p_0)}(\theta, r) \right)^H \tilde{\mathbf{U}}_{n,m} \tilde{\mathbf{U}}_{n,m}^H \mathbf{a}_m^{(p_0)}(\theta, r)}. \tag{41}$$

The spectrum function is evaluated only within the small local grid determined by the coarse estimate $(\hat{\theta}^{(0)}, \hat{r}^{(0)})$.

Accordingly, the local search region is defined as

$$\theta \in [\hat{\theta}^{(0)} - \Delta\theta,\ \hat{\theta}^{(0)} + \Delta\theta], \tag{42}$$

$$r \in [\hat{r}^{(0)} - \Delta r,\ \hat{r}^{(0)} + \Delta r]. \tag{43}$$

This article has been accepted for publication in IEEE Transactions on Vehicular Technology. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVT.2026.3706538

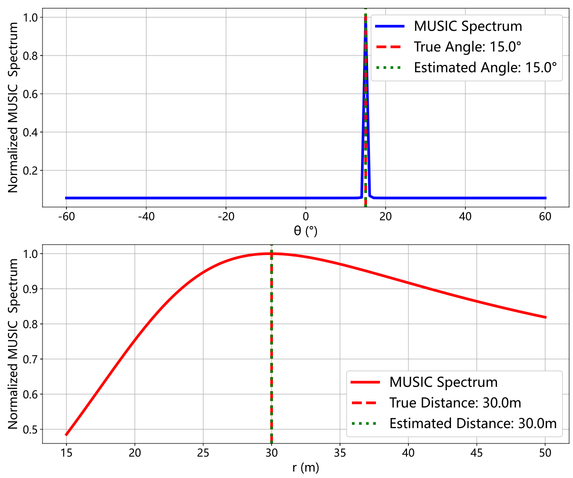


Fig. 6. Example of the two-dimensional near-field MUSIC spatial spectrum in a single-target scenario. A sharp and dominant peak appears at the true target location.

Here, $\Delta\theta$ and Δr define the angular and range search windows used in the localized MUSIC refinement stage. These parameters are selected according to the accuracy of the coarse localization stage. As shown in Fig. 10, the RMSE of the coarse estimation is approximately within 1° in the angular domain and 1 m in the range domain under the considered simulation settings. Therefore, the search window is set to $\Delta\theta = 1^\circ$ and $\Delta r = 1$ m to ensure that the true target location lies within the refinement region with high probability. This choice provides a balance between localization accuracy and computational complexity: a larger search region would increase the computational burden of the MUSIC search, while a smaller region could risk excluding the true target location.

To enhance the robustness of the algorithm in noisy environments, multiple subcarriers around the coarse peak index $m^\star$ are selected, i.e., $m \in \mathcal{S}$, and their local MUSIC spectra are fused through geometric averaging as

$$P_{\text{GMUSIC}}(\theta, r) = \left(\prod_{m \in \mathcal{S}} P_{\text{MUSIC},m}(\theta, r) \right)^{\frac{1}{|\mathcal{S}|}}. \tag{44}$$

The final high-accuracy position estimate is determined by the peak of the fused spectrum function within the local search region as

$$(\hat{\theta}, \hat{r}) = \arg\max_{\theta, r} P_{\text{GMUSIC}}(\theta, r). \tag{45}$$

The uniqueness and correctness of the near-field MUSIC solution are rigorously proven in Appendix A. Compared with the conventional global two-dimensional grid-search-based MUSIC method, the proposed scheme significantly reduces computational complexity through the coarse-to-fine strategy, while the JAD beam pattern design guarantees a unique frequency-space mapping and avoids multi-frequency ambiguity.

Fig. 6 shows an example of the near-field MUSIC spectrum in the single-target case. The coarse estimate provides an initial position around the true target location, and the subsequent localized search produces a distinct dominant peak at the true position, without other comparable peaks in the region of interest. This result verifies the effectiveness of the proposed coarse-to-fine MUSIC refinement and demonstrates its high angular and range resolution.

Algorithm 1 Proposed Coarse-to-Fine Joint Localization Algorithm

Input: Scalar coarse-localization observations $\{z_{k,m}\}$ over all subcarriers, and sequential single-RF observations across all subcarriers and probing intervals.
Output: Final high-accuracy position estimate $(\hat{\theta}_k, \hat{r}_k)$.
// Stage I: Coarse Localization
Compute the received power spectrum $P_{k,m} = |z_{k,m}|^2$ and identify the peak index $m_k^\star = \arg\max_m P_{k,m}$.
Obtain the coarse position estimate $(\hat{\theta}_k^{(0)}, \hat{r}_k^{(0)})$ by inverting the beam-squint-assisted localization formulas.
Define $\hat{\boldsymbol{\eta}}_k^{(0)} = [\hat{\theta}_k^{(0)}, \hat{r}_k^{(0)}]^T$.
// Stage II: Geometry-Compensated Local MUSIC Refinement
Select the neighboring subcarrier set $\mathcal{S}$ around $m_k^\star$.
Divide the full array into P overlapping subarrays of size M_s, and select the central subarray p_0 as the reference subarray.
for each subcarrier $m \in \mathcal{S}$ **do**
  Reconstruct the effective array-level snapshot $\mathbf{y}_{k,m}$ from sequential probing measurements.
  Extract the subarray snapshots $\{\mathbf{y}_{k,m}^{(p)}\}_{p=1}^{P}$.
  For each subarray, construct the near-field phase-alignment matrix $\mathbf{D}_{p,m}(\hat{\boldsymbol{\eta}}_k^{(0)})$ and obtain $\tilde{\mathbf{y}}_{k,m}^{(p)} = \mathbf{D}_{p,m}(\hat{\boldsymbol{\eta}}_k^{(0)})\mathbf{y}_{k,m}^{(p)}$.
  Construct the geometry-compensated covariance matrix $\tilde{\mathbf{R}}_{k,m} = \frac{1}{P}\sum_{p=1}^{P} \tilde{\mathbf{y}}_{k,m}^{(p)} \left(\tilde{\mathbf{y}}_{k,m}^{(p)}\right)^H$.
  Apply EVD to $\tilde{\mathbf{R}}_{k,m}$ and obtain the noise subspace $\tilde{\mathbf{U}}_{n,m}$.
  Evaluate the localized near-field MUSIC spectrum $P_{\text{MUSIC},m}(\theta, r)$ over the local grid $\mathcal{G}$ using the reference-subarray steering vector $\mathbf{a}_m^{(p_0)}(\theta, r)$.
end for
// Stage III: Spectrum Fusion and Output
Fuse $\{P_{\text{MUSIC},m}(\theta, r)\}_{m \in \mathcal{S}}$ by geometric averaging to obtain $P_{\text{GMUSIC}}(\theta, r)$.
Identify the peak of the fused spectrum within $\mathcal{G}$: $(\hat{\theta}_k, \hat{r}_k) = \arg\max_{(\theta,r)\in\mathcal{G}} P_{\text{GMUSIC}}(\theta, r)$.
return $(\hat{\theta}_k, \hat{r}_k)$.

To clearly illustrate the proposed framework, we summarize the complete procedure in Algorithm 1, which integrates the efficiency of coarse estimation with the super-resolution capability of MUSIC-based fine estimation.

2) Multi-Target Joint Localization: In practical communication and sensing scenarios, the BS often needs to simultaneously serve and sense multiple users, making multi-target joint localization a critical metric for evaluating the practicality of any algorithm. With the aid of the designed JAD localization, each user corresponds to a unique peak subcarrier index $m_k^\star$ in the frequency domain, thereby enabling user separation and

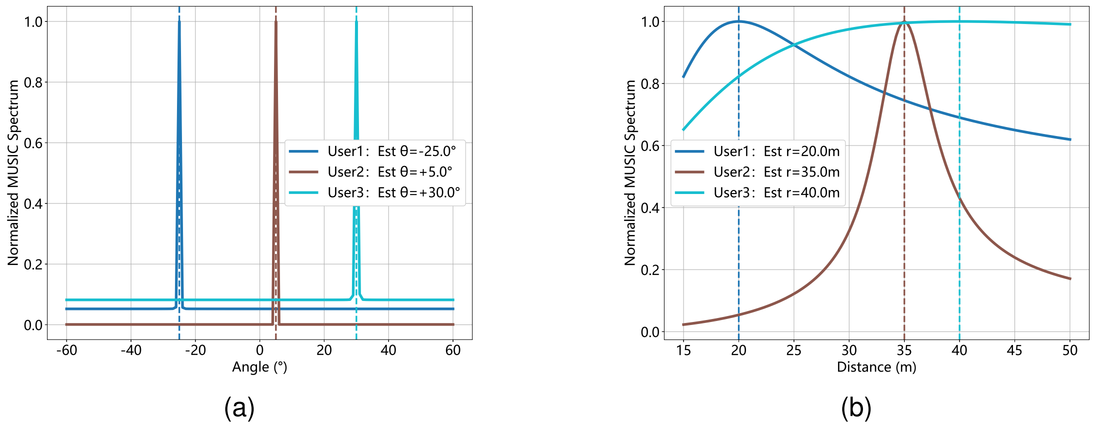


Fig. 7. Joint localization results in Multi-User scenario. (a) Normalized MUSIC power spectrum in angular dimension. (b) Normalized MUSIC power spectrum in range dimension.

coarse localization in the first stage.

In the refinement stage, the near-field MUSIC algorithm constructs the noise subspace and leverages its orthogonality to the true steering vectors, allowing the spectral peaks corresponding to different users to be clearly separated in the two-dimensional angle-distance domain. Thanks to its super-resolution property, MUSIC can effectively suppress mutual interference and maintain high resolution even when users are closely spaced in angle and distance.

To further evaluate the proposed method in the multi-user case, Fig. 7 presents the joint localization results under the default simulation configuration summarized in Table II. Unless otherwise specified, the same wideband XL-MIMO setting and MUSIC refinement parameters are adopted here. It can be observed that sharp and prominent peaks appear precisely at the true locations of all users, while no spurious peaks emerge elsewhere. This result demonstrates that the proposed method not only achieves extremely high accuracy in single-user cases but also delivers outstanding localization performance in multi-user environments.

3) Extension to Uniform Planar Array (UPA): Although the proposed framework is introduced based on a ULA for clarity, it can be naturally extended to a uniform planar array (UPA) to enable three-dimensional near-field localization. Consider an $N_x \times N_y$ UPA deployed on the x-y plane with inter-element spacing d. The coordinate of the (n_x, n_y)-th antenna element is

$$(x_{n_x}, y_{n_y}) = (n_x d,\, n_y d), \tag{46}$$

where $n_x = -\frac{N_x-1}{2}, \ldots, \frac{N_x-1}{2}$ and $n_y = -\frac{N_y-1}{2}, \ldots, \frac{N_y-1}{2}$.

For a user located at (r, θ, ϕ), where θ and ϕ denote the elevation and azimuth angles, respectively, the distance between the user and the (n_x, n_y)-th antenna element is given by

$$\begin{aligned} &r_{n_x,n_y} \\ &= \sqrt{r^2 + x_{n_x}^2 + y_{n_y}^2 - 2r(x_{n_x}\sin\theta\cos\phi + y_{n_y}\sin\theta\sin\phi)}. \end{aligned} \tag{47}$$

Accordingly, the near-field steering vector can be written as

$$\mathbf{a}(r,\theta,\phi,f_m) = \frac{1}{\sqrt{N_x N_y}}\left[e^{-j\frac{2\pi f_m}{c} r_{n_x,n_y}}\right]. \tag{48}$$

In the coarse localization stage, the beam-squint-induced frequency-space mapping is generalized from the ULA case to a three-dimensional trajectory. Defining the direction-cosine variables as

$$u_x = \sin\theta\cos\phi, \qquad u_y = \sin\theta\sin\phi, \tag{49}$$

the peak subcarrier index

$$m^\star = \arg\max_m |z_m|^2 \tag{50}$$

can be mapped to an initial estimate $(\hat{\phi}^{(0)}, \hat{\theta}^{(0)}, \hat{r}^{(0)})$ through the corresponding UPA beam-squint trajectory.

In the fine localization stage, the MUSIC spectrum is extended to

$$P_{\text{MUSIC}}(\phi,\theta,r) = \frac{1}{\mathbf{a}^H(\phi,\theta,r)\mathbf{U}_n\mathbf{U}_n^H\mathbf{a}(\phi,\theta,r)}, \tag{51}$$

which jointly resolves the azimuth, elevation, and range of the user. Therefore, the proposed coarse-to-fine framework can be directly extended from ULA-based two-dimensional localization to UPA-based three-dimensional near-field localization.

To illustrate this extension, we consider a 16×16 UPA operating at $f_c = 60$ GHz with inter-element spacing $d = \lambda/2$, where the true user position is $(\phi_0, \theta_0, r_0) = (20^\circ, 30^\circ, 25\ \mathrm{m})$. Fig. 8 (a) shows the azimuth-elevation slice of the MUSIC spectrum at the true range $r = r_0$, while Fig. 8 (b) shows the range-domain cut at the true angular position $(\phi, \theta) = (\phi_0, \theta_0)$. In both cases, a clear dominant peak appears at the true user location, which verifies the effectiveness of the proposed UPA-based extension.

D. Computational Complexity Analysis

The computational complexity of the proposed coarse-to-fine localization algorithm is mainly dominated by four steps: 1) the coarse power-spectrum evaluation over all subcarriers; 2) the construction of the geometry-compensated covariance

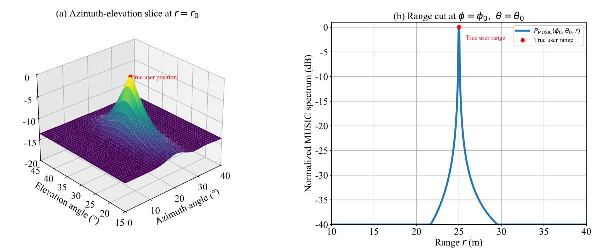


Fig. 8. near-field MUSIC spectrum under the UPA setting for a user located at $(\phi_0, \theta_0, r_0) = (20^\circ, 30^\circ, 25\text{ m})$. (a) Azimuth-elevation slice at the true user range $r = r_0$. (b) Range-domain cut at the true azimuth and elevation angles $(\phi, \theta) = (\phi_0, \theta_0)$.

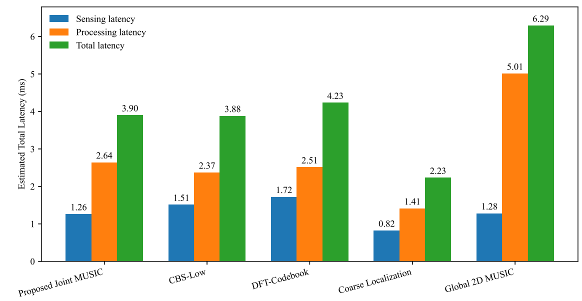


Fig. 9. Estimated sensing, processing, and total latency comparison of different localization schemes.

TABLE I
COMPLEXITY COMPARISON

Method	Computational Complexity
CBS-Low [20]	$\mathcal{O}(M)$
Proposed method	$\mathcal{O}\big(M + \lvert\mathcal{S}\rvert(PM_s^2 + M_s^3 + N_g^{\text{loc}}M_s^2)\big)$
Global 2D MUSIC	$\mathcal{O}\big(\lvert\mathcal{S}\rvert(M_s^3 + N_g P M_s^2)\big)$

matrices, including near-field phase alignment; 3) the eigenvalue decomposition (EVD) of the geometry-compensated covariance matrices; and 4) the localized near-field MUSIC spectrum evaluation. Specifically, the complexity of the coarse localization stage is $\mathcal{O}(M)$, where M denotes the total number of subcarriers. For the refinement stage, let $|\mathcal{S}|$ denote the number of selected subcarriers used for local refinement, P the number of overlapping subarrays, M_s the subarray size, and $N_g^{\text{loc}} = N_\theta^{\text{loc}} N_r^{\text{loc}}$ the number of local grid points. The additional near-field phase-alignment operation requires $\mathcal{O}(|\mathcal{S}|PM_s)$ operations, which is negligible compared with the covariance construction, EVD, and MUSIC spectrum evaluation. Therefore, the dominant complexity of the proposed method can be written as

$$\mathcal{O}\big(M + |\mathcal{S}|\big(PM_s^2 + M_s^3 + N_g^{\text{loc}}M_s^2\big)\big).$$

For comparison, the CBS-Low method mainly relies on subcarrier power-spectrum peak detection and closed-form parameter inversion, and therefore its dominant complexity is $\mathcal{O}(M)$. By contrast, the conventional global two-dimensional MUSIC method performs the spectrum search over the full angle-range grid, whose dominant complexity is

$$\mathcal{O}\big(|\mathcal{S}|\big(PM_s^2 + M_s^3 + N_g M_s^2\big)\big),$$

where $N_g = N_\theta N_r$ denotes the number of global grid points. Since $N_g^{\text{loc}} \ll N_g$ in practice, the proposed method significantly reduces the computational burden compared with the global two-dimensional MUSIC search, while maintaining substantially better localization accuracy than CBS-Low. The dominant complexity comparison is summarized in Table I.

It should be noted that the DFT-codebook and Coarse Estimation baselines are not included in Table I. The reason is that these conventional search/lookup or peak-detection-based schemes do not have a unique FLOP-order expression independent of the codebook size, search granularity, and implementation strategy. Therefore, to provide a more intuitive comparison among all online schemes, we further evaluate the estimated sensing, processing, and total latency under unified assumptions.

TABLE II
DEFAULT SIMULATION CONFIGURATION

Parameter	Value
f_c	60 GHz
N_t	256
Array architecture	ULA
B	3 GHz
M	2048
d	$\lambda/2$
M_s	128
P	$N_t - M_s + 1 = 129$
Sensing region	$-60^\circ \le \theta \le 60^\circ$, $15\text{ m} \le r \le 50\text{ m}$

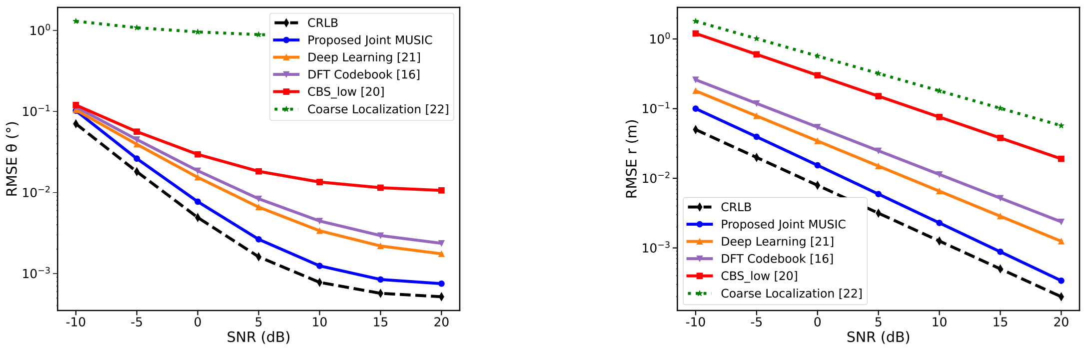
As shown in Fig. 9, the proposed method has a total latency of 3.90 ms, which is almost the same as CBS-Low (3.88 ms), but it achieves much higher localization accuracy due to the proposed localized near-field MUSIC refinement. Compared with DFT-Codebook, the proposed method also requires lower latency (3.90 ms versus 4.23 ms), while avoiding broad codebook searching. Global 2D MUSIC has the largest latency because of its full angle-range grid search, whereas Coarse Estimation has the lowest latency but much poorer accuracy due to the lack of refinement. Therefore, the proposed method provides a better trade-off between latency and localization accuracy.

V. SIMULATION RESULTS

In this section, simulation results are presented to validate the effectiveness of the proposed beam-squint-assisted JAD localization scheme. Unless otherwise specified, all simulations adopt the default system configuration summarized in Table II. Specifically, we consider a wideband XL-MIMO system operating at a carrier frequency of $f_c = 60$ GHz, where the BS is equipped with a ULA of $N_t = 256$ antennas and employs a TTD-aided beamforming architecture. The system bandwidth is set to $B = 3$ GHz and equally divided into $M = 2048$ subcarriers. The inter-element spacing is $d = \lambda/2$, and the noise is modeled as complex Gaussian with variance σ^2.

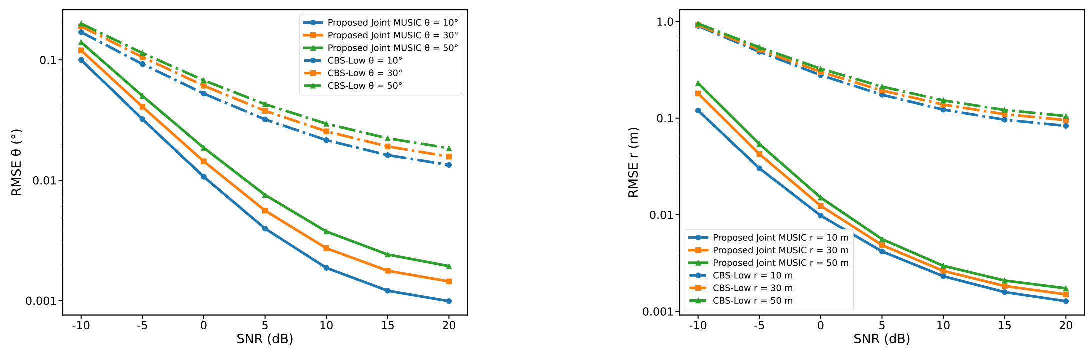
For the MUSIC-based refinement stage, near-field geometry-compensated spatial smoothing is performed

This article has been accepted for publication in IEEE Transactions on Vehicular Technology. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/TVT.2026.3706538



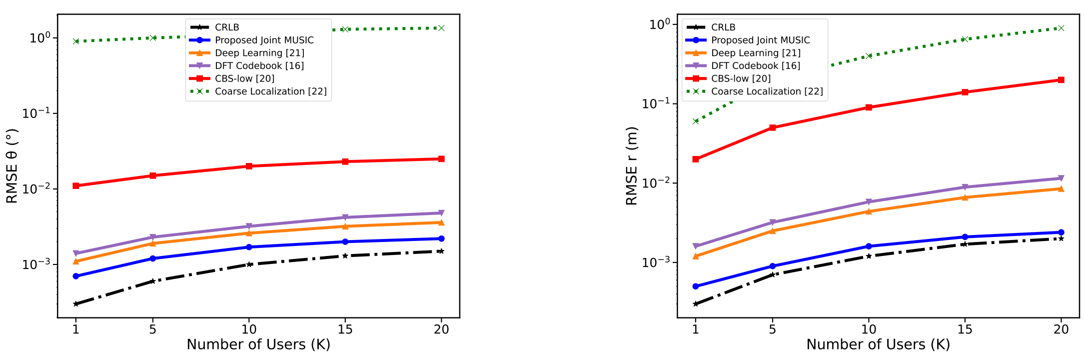
(a) (b)

Fig. 10. (a) Angle estimation RMSE versus SNR for different localization schemes. (b) Distance estimation RMSE versus SNR for different localization schemes. Note that different schemes employ different sensing and refinement mechanisms.



(a) (b)

Fig. 11. (a) Angle estimation RMSE of the proposed method. (b) Distance estimation RMSE of the proposed method. Different curves represent users at different locations.



(a) (b)

Fig. 12. (a) Angle estimation RMSE versus the number of users in the multi-user scenario. (b) Distance estimation RMSE versus the number of users in the multi-user scenario. Different curves represent different methods.

by partitioning the full array into overlapping subarrays and phase-aligning the subarray snapshots to the central reference subarray. The subarray size is set to $M_s = 128$. With unit shifting between adjacent subarrays, the number of overlapping subarrays is $P = N_t - M_s + 1 = 256 - 128 + 1 = 129$. This setting provides a practical trade-off among covariance reconstruction, subspace estimation accuracy, and computational complexity.

Both single-user and multi-user near-field scenarios are considered to comprehensively evaluate the proposed scheme in terms of localization accuracy, robustness, and computational efficiency. For performance comparison, we consider the following benchmark schemes.

- **Two-Step Localization (CBS-Low)** [20]: This method

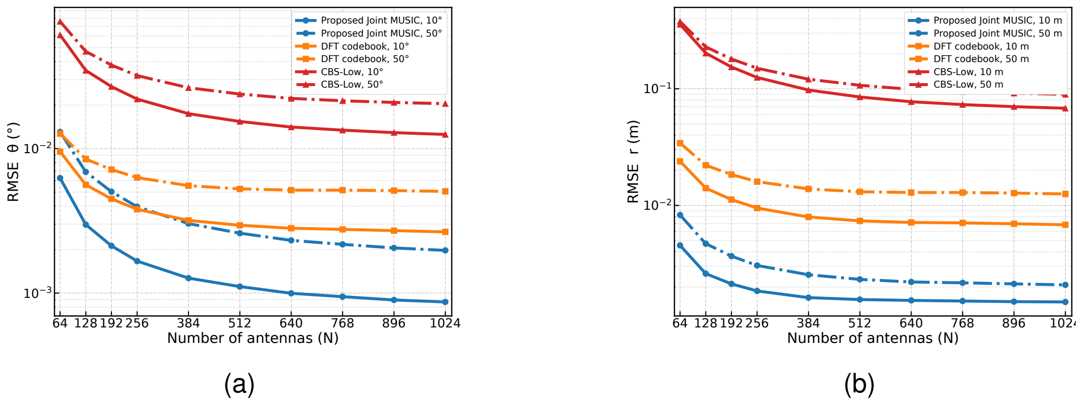


Fig. 13. Localization RMSE versus the number of antennas under fixed SNR = 10 dB. (a) Angle estimation RMSE under different user angles. (b) Distance estimation RMSE under different user ranges.

is a representative controllable beam-squint baseline that estimates the angle first and then the distance in a cascaded manner.

- **Deep Learning-Based Approach** [21]: This method is a recent learning-based near-field localization benchmark that combines controllable beam squint with neural-network-based refinement.
- **DFT Codebook-Based Approach** [16]: This method is a codebook-based near-field beam training scheme for joint angle and range estimation.
- **Coarse Localization** [22]: This method is a coarse localization baseline that performs cascaded angle and range estimation without the proposed localized refinement.
- **Cramér-Rao Lower Bound (CRLB)**: The CRLB serves as the theoretical lower bound on the RMSE of unbiased estimators under the assumed signal and noise model.

A. Single-User Scenario

Fig. 10 (a) shows the angle RMSE versus SNR. As the SNR increases, the RMSE of all methods decreases, while the proposed method consistently achieves the lowest error among all practical schemes and remains closest to the CRLB. The deep-learning and DFT-codebook baselines also provide relatively good performance, but they are still inferior to the proposed method due to model mismatch and codebook resolution limitations. By contrast, CBS-Low and Coarse Estimation degrade more noticeably, since their angle estimates rely more heavily on peak detection and coarse spectral responses, making them more sensitive to noise and quantization effects.

Fig. 10 (b) shows the distance RMSE versus SNR. The proposed method again achieves the lowest RMSE among all practical schemes and remains nearest to the CRLB, with the advantage being especially clear in the low-to-medium SNR region. This is because the proposed framework jointly exploits the near-field quadratic phase and performs localized two-dimensional refinement, which effectively suppresses the range degradation caused by angular-distance coupling. The deep-learning and DFT-codebook baselines still outperform CBS-Low and Coarse Estimation, but their distance errors remain higher due to limited refinement capability. In comparison, CBS-Low suffers from sequential error propagation, while Coarse Estimation lacks the high-resolution refinement stage, leading to faster performance deterioration under noise.

B. Multi-User Scenario

Fig. 11 (a) compares the angle estimation RMSE versus SNR for the proposed method and the CBS-Low scheme in the multi-user scenario, where different users are located at different angular positions, namely $\theta = 10°$, $30°$, and $50°$. It can be observed that the proposed method consistently achieves lower angle estimation RMSE than CBS-Low over the entire tested SNR range for all considered user angles. Moreover, the angle estimation performance of both methods degrades as the user angle increases. Nevertheless, the proposed method maintains a clear performance advantage over CBS-Low across all considered angular positions, indicating better robustness under angular diversity in the multi-user scenario.

Fig. 11 (b) compares the distance estimation RMSE versus SNR for the proposed method and the CBS-Low scheme in the multi-user scenario, where different users are located at different ranges, namely $r = 10$ m, 30 m, and 50 m. It is clear that the proposed method consistently outperforms CBS-Low for all considered user distances and SNR values. As the user distance increases, the distance estimation RMSE of both methods also increases. However, the proposed method still preserves lower RMSE than CBS-Low throughout the entire SNR range, which demonstrates stronger robustness under range diversity in the multi-user near-field scenario.

Fig. 12 (a) shows the angle RMSE versus user density. As the number of users increases from sparse to dense regimes, the angle RMSE of all practical methods gradually increases due to stronger multi-user coupling. Nevertheless, the proposed joint localization algorithm consistently achieves the lowest angular RMSE among all benchmark schemes and remains closest to the CRLB. The deep-learning and DFT-codebook baselines also show relatively good robustness, while CBS-Low and Coarse Estimation degrade more noticeably as user density increases.

This article has been accepted for publication in IEEE Transactions on Vehicular Technology. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVT.2026.3706538

Fig. 12 (b) shows the distance RMSE versus user density. As user number grows, the range RMSE of all practical methods increases because the radial-domain separability becomes weaker under stronger interference. Even so, the proposed method maintains the lowest RMSE among all benchmark schemes and remains nearest to the CRLB over the whole range. The deep-learning and DFT-codebook baselines still outperform CBS-Low and Coarse Estimation, but their degradation becomes more evident in high-density regimes. These results demonstrate that the proposed JAD framework provides superior robustness, scalability, and localization reliability for dense near-field deployments.

C. Impact of Antenna Number

To further evaluate the influence of array size on near-field localization, we compare the angle and distance RMSE of different methods when the number of antennas increases from 64 to 1024. The SNR is fixed at 10 dB, and the remaining system parameters follow the default simulation setup unless otherwise specified.

Fig. 13 (a) shows the angle RMSE versus the number of antennas. As N increases, all methods achieve lower RMSE due to the enlarged aperture and improved angular resolution. The proposed method consistently outperforms DFT-codebook and CBS-Low for both $\theta = 10^\circ$ and $\theta = 50^\circ$. This advantage comes from the localized near-field MUSIC refinement, which better exploits the aperture gain. By contrast, DFT-codebook is constrained by finite codebook resolution, while CBS-Low is affected by the cascaded angle-first-then-distance estimation structure. The higher RMSE at $\theta = 50^\circ$ also indicates that larger angular offsets are more challenging.

Fig. 13 (b) shows the distance RMSE versus the number of antennas. The distance RMSE decreases with N, because the larger aperture strengthens the distance-dependent quadratic phase variation in the near-field steering vector. The proposed method maintains the lowest RMSE over the whole antenna range. The DFT-codebook method improves more slowly due to codebook limitations, while CBS-Low suffers from error propagation from the angle estimation stage. The $r = 50$ m case remains more difficult than $r = 10$ m because the near-field curvature becomes weaker at longer distances. These results verify the robustness of the proposed method under different array sizes.

VI. Conclusion

In this paper, we investigated the problem of high-accuracy user localization in near-field wideband XL-MIMO systems. We first analyzed the near-field beam-squint effect and designed a beam-squint-assisted JAD localization scheme that effectively covers the two-dimensional sensing region. Based on this design, we proposed a coarse-to-fine two-stage localization framework. In the first stage, a coarse estimate of the target position is obtained from the received power spectrum. In the second stage, a localized near-field MUSIC refinement is performed around the coarse estimate to achieve high-resolution joint estimation of angle and distance. By avoiding conventional sequential angle-first-then-distance processing, the proposed framework mitigates error propagation and improves localization accuracy. Simulation results in both single-user and multi-user scenarios verified the accuracy, robustness, and applicability of the proposed method for near-field wideband XL-MIMO localization.

Appendix A
Proof of the Uniqueness of the Near-Field MUSIC Solution

This section aims to provide a theoretical proof that, for the proposed MUSIC algorithm based on near-field steering vectors, the peak of the spatial spectrum is unique, and the spatial coordinate (θ, r) corresponding to this peak precisely represents the true position of the target.

The proof consists of two steps: first, we show that at the true target location, the spectrum function exhibits a peak (correctness); second, we demonstrate that at any non-true location, no peak occurs (uniqueness).

A. Proof of correctness

According to the monostatic sensing model, the ideal signal covariance matrix is given by $R = P_s a(\theta_k, r_k) a(\theta_k, r_k)^H + \sigma_n^2 I$.where $a(\theta_k, r_k)$ denotes the steering vector. By performing eigenvalue decomposition (EVD) on R, the signal subspace $\mathcal{U}_s$ is spanned by the true steering vector $a(\theta_k, r_k)$, while the noise subspace $\mathcal{U}_n$ is orthogonal to $\mathcal{U}_s$.

This implies that the true steering vector $a(\theta_k, r_k)$ must be orthogonal to any basis vector of the noise subspace $\mathcal{U}_n$. Let U_n denote the matrix composed of all basis vectors of $\mathcal{U}_n$, then it follows that

$$a(\theta_k, r_k)^H U_n = \mathbf{0}. \tag{52}$$

The MUSIC pseudo-spectrum function is defined as

$$P_{\text{MUSIC}}(\theta, r) = \frac{1}{a(\theta, r)^H U_n U_n^H a(\theta, r)}. \tag{53}$$

When the true target location (θ_k, r_k) is substituted into the above expression, the denominator becomes zero according to (52).

In the ideal case, this leads to $P_{\text{MUSIC}}(\theta_k, r_k) \to \infty$. This proves that at the true target location, the MUSIC spectrum function necessarily exhibits a peak, which validates the correctness of the solution.

B. Proof of uniqueness

To establish uniqueness, we must prove that for any “false” location $(\theta', r') \neq (\theta_k, r_k)$, the spectrum value $P_{\text{MUSIC}}(\theta', r')$ cannot form a peak, i.e., its denominator is strictly non-zero.

We proceed by contradiction. Suppose that a false location (θ', r') also yields a peak. Then its denominator must also vanish, which implies that $a(\theta', r')$ is orthogonal to the noise subspace. In the single-target case, the signal subspace is one-dimensional, hence $a(\theta', r')$ must be linearly dependent on the true steering vector $a(\theta_k, r_k)$:

$$a(\theta', r') = c \cdot a(\theta_k, r_k), \tag{54}$$

where c is a nonzero complex scalar.

This equivalently requires that their phase difference is constant across all antennas n.

$$\Phi_n(\theta', r') - \Phi_n(\theta_k, r_k) = \text{const}, \quad \forall n. \tag{55}$$

By substituting the phase expressions and simplifying, we obtain

$$\underbrace{(r_k - r')}_{C_0} + n_d \cdot \underbrace{d(\sin\theta' - \sin\theta_k)}_{C_1} + n_d^2 \cdot \underbrace{\frac{d^2}{2}\left(\frac{\cos^2\theta_k}{r_k} - \frac{\cos^2\theta'}{r'}\right)}_{C_2} = \text{const}'. \tag{56}$$

This expression forms a quadratic polynomial in the antenna index n_d. For this polynomial to remain constant across all N distinct values of n_d, both the linear and quadratic coefficients must vanish simultaneously (for arrays with $N \geq 3$).

Therefore, the only condition under which (56) holds is $(\theta', r') = (\theta_k, r_k)$. This contradicts our initial assumption that a false location $(\theta', r') \neq (\theta_k, r_k)$ could also generate a peak.

Thus, the assumption is invalid. No "ambiguous point" distinct from the true location can produce a steering vector that is linearly related to the true one. This implies that for any $(\theta', r') \neq (\theta_k, r_k)$, the steering vector $a(\theta', r')$ must be non-orthogonal to the noise subspace, and the denominator of the MUSIC spectrum remains strictly positive, preventing any peak formation.This completes the proof of the uniqueness of the solution.

Appendix B
Derivation of the CRLB for the Proposed Near-Field Localization Model

The Cramér-Rao lower bound (CRLB) provides a theoretical lower limit on the mean squared error of any unbiased estimator and serves as a fundamental benchmark for evaluating the statistical efficiency of the proposed estimation algorithm. In this section, we derive the CRLB for joint angle and distance estimation in a monostatic, single-target, near-field sensing scenario.

Assume that the base station is equipped with a uniform linear array (ULA) of N elements and receives L snapshots of echo signals from a single target located at (θ, r). At the l-th snapshot, the received signal vector $y_l \in \mathbb{C}^{N\times 1}$ can be modeled as

$$y_l = a(\theta, r)s_l + n_l, \quad l = 1, \ldots, L, \tag{57}$$

where s_l is the complex reflection coefficient of the l-th snapshot (accounting for path loss and target RCS), and $n_l \sim \mathcal{CN}(0, \sigma^2 I)$ denotes zero-mean circularly symmetric complex Gaussian noise. The vector $a(\theta, r)$ represents the near-field steering vector, whose n-th element is given as

$$\begin{aligned}[a(\theta, r)]_n &= \exp\{j\Phi_n(\theta, r)\} \\ &= \exp\left\{-j\frac{2\pi f_c}{c}\left(r - n_d d\sin\theta + \frac{(n_d d)^2\cos^2\theta}{2r}\right)\right\},\end{aligned} \tag{58}$$

where $n_d = n - (N-1)/2$ denoting the antenna index referenced to the array center.

The unknown parameters to be estimated are the angle θ and distance r. In practice, the reflection coefficients s_l and noise variance σ^2 may also be unknown. For analytical tractability, we consider a more general case in which the complex amplitudes s_l are treated as unknown deterministic parameters. The real-valued parameter vector to be estimated is then defined as $\boldsymbol{\eta} = [\theta, r]^T$.

For the above signal model, the Fisher information matrix (FIM) of the parameter vector $\boldsymbol{\eta}$, denoted by $J(\boldsymbol{\eta})$, is a 2×2 matrix.In the presence of complex Gaussian noise, its (i,j)-th entry can be computed as

$$[J(\boldsymbol{\eta})]_{i,j} = \frac{2}{\sigma^2}\sum_{l=1}^{L}|s_l|^2 \cdot \mathrm{Re}\left\{\left(\frac{\partial a}{\partial\eta_i}\right)^H\left(\frac{\partial a}{\partial\eta_j}\right)\right\}, \tag{59}$$

where $\eta_1 = \theta$, $\eta_2 = r$, and $\frac{\partial a}{\partial \eta_i}$ denotes the partial derivative of the steering vector with respect to the parameter η_i.

First, we compute the partial derivatives of the steering vector with respect to θ and r.Let $d_\theta = \frac{\partial a(\theta,r)}{\partial\theta}$ and $d_r = \frac{\partial a(\theta,r)}{\partial r}$.Their n-th elements are given as

$$[d_\theta]_n = [a(\theta,r)]_n \cdot j\frac{\partial\Phi_n(\theta,r)}{\partial\theta}, \tag{60}$$

$$[d_r]_n = [a(\theta,r)]_n \cdot j\frac{\partial\Phi_n(\theta,r)}{\partial r}, \tag{61}$$

where the partial derivatives of the phase are given as

$$\begin{aligned}\frac{\partial\Phi_n(\theta,r)}{\partial\theta} &= -\frac{2\pi f_c}{c}\left(-n_d d\cos\theta - \frac{(n_d d)^2\cdot 2\cos\theta(-\sin\theta)}{2r}\right) \\ &= \frac{2\pi f_c}{c}\left(n_d d\cos\theta - \frac{(n_d d)^2\sin(2\theta)}{2r}\right),\end{aligned} \tag{62}$$

$$\frac{\partial\Phi_n(\theta,r)}{\partial r} = -\frac{2\pi f_c}{c}\left(1 - \frac{(n_d d)^2\cos^2\theta}{2r^2}\right). \tag{63}$$

The total signal-to-noise ratio (SNR) is defined as $\mathrm{SNR} = \frac{L\sum|s_l|^2}{N\sigma^2}$.For simplicity, we consider the single-snapshot case $L = 1$, and define the effective SNR with normalized signal power as $\mathrm{SNR} = \frac{L\sum|s_l|^2}{N\sigma^2}$.The FIM can then be expressed as

$$\begin{aligned}J &= \frac{2|s|^2 L}{\sigma^2}\begin{pmatrix}\mathrm{Re}\{d_\theta^H d_\theta\} & \mathrm{Re}\{d_\theta^H d_r\} \\ \mathrm{Re}\{d_r^H d_\theta\} & \mathrm{Re}\{d_r^H d_r\}\end{pmatrix} \\ &= 2\mathrm{SNR}_{\text{eff}} L\begin{pmatrix}\|d_\theta\|^2 & \mathrm{Re}\{d_\theta^H d_r\} \\ \mathrm{Re}\{d_r^H d_\theta\} & \|d_r\|^2\end{pmatrix},\end{aligned} \tag{64}$$

where $|d_\theta|^2 = \sum_{n=1}^{N}\left|\frac{\partial\Phi_n}{\partial\theta}\right|^2$ and $|d_r|^2 = \sum_{n=1}^{N}\left|\frac{\partial\Phi_n}{\partial r}\right|^2$.

The CRLB is given by the inverse of the FIM. For a 2×2 matrix $J = \begin{pmatrix}A & B \\ B & C\end{pmatrix}$, the inverse is $J^{-1} = \frac{1}{AC-B^2}\begin{pmatrix}C & -B \\ -B & A\end{pmatrix}$.

Therefore, the CRLBs for angle and distance estimation are given as

$$\begin{aligned}\text{CRLB}(\theta) &= [J^{-1}]_{1,1} = \frac{J_{2,2}}{\det(J)} \\ &= \frac{\|d_r\|^2}{2\text{SNR}_{\text{eff}}L\left(\|d_\theta\|^2\|d_r\|^2 - (\text{Re}\{d_\theta^H d_r\})^2\right)},\end{aligned} \tag{65}$$

$$\begin{aligned}\text{CRLB}(r) &= [J^{-1}]_{2,2} = \frac{J_{1,1}}{\det(J)} \\ &= \frac{\|d_\theta\|^2}{2\text{SNR}_{\text{eff}}L\left(\|d_\theta\|^2\|d_r\|^2 - (\text{Re}\{d_\theta^H d_r\})^2\right)}.\end{aligned} \tag{66}$$

The lower bound of the root-mean-square error (RMSE) is given by $\text{RMSE} \geq \sqrt{\text{CRLB}}$. Equations (65) and (66) serve as the theoretical performance bounds adopted in the simulations of this paper.

REFERENCES

[1] M. Cui, Z. Wu, Y. Lu, X. Wei, and L. Dai, "Near-field MIMO Communications for 6G: Fundamentals, Challenges, Potentials, and Future Directions," *IEEE Commun. Mag.*, vol. 61, no. 1, pp. 40–46, 2022.

[2] J. Lu, J. Zhang, H. Lei, H. Xiao, B. Ai, and D. W. Kwan Ng, "Near-Field Spatial-Domain Channel Extrapolation for XL-MIMO Systems," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 4343–4357, 2026.

[3] C. Ruan, Z. Zhang, H. Jiang, J. Dang, L. Wu, and H. Zhang, "Wideband near-field channel covariance estimation for XL-MIMO systems in the face of beam split," *IEEE Trans. Veh. Technol.*, 2024.

[4] R. Zhang, L. Cheng, S. Wang, Y. Lou, Y. Gao, W. Wu, and D. W. K. Ng, "Integrated Sensing and Communication With Massive MIMO: A Unified Tensor Approach for Channel and Target Parameter Estimation," *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, pp. 8571–8587, 2024.

[5] R. Zhang, G. Chen, L. Cheng, X. Guan, Q. Wu, W. Wu, and R. Zhang, "Tensor-Based Channel Estimation for Extremely Large-Scale MIMO-OFDM With Dynamic Metasurface Antennas," *IEEE Trans. Wireless Commun.*, vol. 24, no. 7, pp. 6052–6068, 2025.

[6] Q. Zhang, H. Wang, X. Shen, Y. Yan, and Z. Zhao, "Enhancing Underwater DOA Estimation Accuracy With Limited Datasets Using Task-Restructured Deep Mutual Learning," *IEEE Signal Process. Lett.*, vol. 32, pp. 966–970, 2025.

[7] R. Schmidt, "Multiple Emitter Location and Signal Parameter Estimation," *IEEE Trans. Antennas Propag.*, vol. 34, no. 3, pp. 276–280, 1986.

[8] Q. Yang, A. Guerra, F. Guidi, N. Shlezinger, H. Zhang, D. Dardari, B. Wang, and Y. C. Eldar, "Beam Focusing for Near-Field Multi-User Localization," *IEEE Trans. Veh. Technol.*, 2025.

[9] M. Cui and L. Dai, "Channel Estimation for Extremely Large-Scale MIMO: Far-Field or Near-Field?" *IEEE Trans. Commun.*, vol. 70, no. 4, pp. 2663–2677, 2022.

[10] S. Zhang, Y. Zhang, Y. Liu, and B. Di, "Codebook Design and Beam Alignment for IOS-Aided Communications: From the Near-Field and Far-Field Boundary Perspective," *IEEE Trans. Wireless Commun.*, 2025.

[11] H. Zhang, N. Shlezinger, F. Guidi, D. Dardari, M. F. Imani, and Y. C. Eldar, "Beam Focusing for Near-Field Multiuser MIMO Communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 7476–7490, 2022.

[12] C. Huang, J. Xu, W. Xu, X. You, C. Yuen, and Y. Chen, "Low-complexity channel estimation for extremely large-scale MIMO in near field," *IEEE Wireless Commun. Lett.*, vol. 13, no. 3, pp. 671–675, 2023.

[13] H. Hua, J. Xu, and R. Zhang, "Near-Field Integrated Sensing and Communication with Extremely Large-Scale Antenna Array," *IEEE Trans. Wireless Commun.*, pp. 1–1, 2025.

[14] L. Khamidullina, I. Podkurkov, and M. Haardt, "Conditional and Unconditional Cramér-Rao bounds for Near-Field Localization in Bistatic MIMO Radar Systems," *IEEE Trans. Signal Process.*, vol. 69, pp. 3220–3234, 2021.

[15] A. Guerra, F. Guidi, D. Dardari, and P. M. Djurić, "Near-Field Tracking with Large Antenna Arrays: Fundamental Limits and Practical Algorithms," *IEEE Trans. Signal Process.*, vol. 69, pp. 5723–5738, 2021.

[16] X. Wu, C. You, J. Li, and Y. Zhang, "Near-field beam training: Joint angle and range estimation with DFT codebook," *IEEE Trans. Wireless Commun.*, vol. 23, no. 9, pp. 11 890–11 903, 2024.

[17] W. Hao, X. You, F. Zhou, Z. Chu, G. Sun, and P. Xiao, "The Far-/Near-Field Beam Squint and Solutions for THz Intelligent Reflecting Surface Communications," *IEEE Trans. Veh. Technol.*, vol. 72, no. 8, pp. 10 107–10 118, 2023.

[18] M. Abuyaghi, S. Si-Mohammed, G. Shaker, and C. Rosenberg, "Positioning in 5G Networks: Emerging Techniques, Use Cases, and Challenges," *IEEE Internet Things J.*, vol. 12, no. 2, pp. 1408–1427, 2025.

[19] G. Zhou, M. Garkisch, Z. Peng, C. Pan, and R. Schober, "Radar Rainbow Beams for Wideband mmWave Communication: Beam Training and Tracking," *IEEE J. Sel. Areas Commun.*, 2025.

[20] H. Luo, F. Gao, W. Yuan, and S. Zhang, "Beam Squint Assisted User Localization in Near-Field Integrated Sensing and Communications Systems," *IEEE Trans. Wireless Commun.*, vol. 23, no. 5, pp. 4504–4517, 2023.

[21] H. Lei, J. Zhang, H. Xiao, D. W. K. Ng, and B. Ai, "Deep Learning Based Near-Field User Localization with Beam Squint in Wideband XL-MIMO Systems," *IEEE Trans. Wireless Commun.*, 2024.

[22] A. Gast, L. Le Magoarou, and N. Shlezinger, "DCD-MUSIC: Deep-Learning-Aided Cascaded Differentiable MUSIC Algorithm for Near-Field Localization of Multiple Sources," in *Proc. IEEE ICASSP*, 2025, pp. 1–5.